Web Application Software Architecture:

Web application architecture defines the structure and interactions between components in a web application. Let's explore the key concepts:

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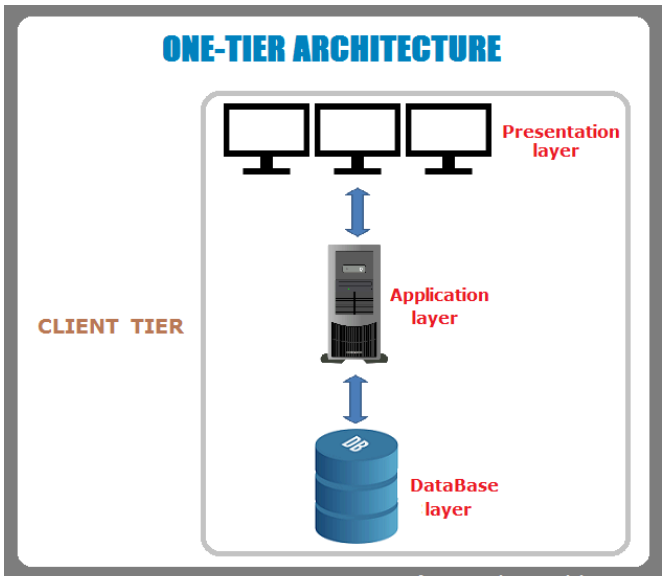
1. Software Architecture Types

1.1 Single-Tier Architecture

In a **single-tier architecture**, the application's entire logic resides in one layer, typically on the client machine. It is simple but not scalable for larger applications.



Example:

- A desktop application where the database and user interface exist in the same system (e.g., Microsoft Access).

Advantages:

- Simple development and deployment.
- Fast data access as everything is on the same machine.

Disadvantages:

- Limited scalability.
- Difficult to maintain and update.

1.2 Two-Tier Architecture

In a **two-tier architecture**, the client and server are separated:

1. **Client Tier**: Contains the user interface.
2. **Server Tier**: Manages the database and business logic.

Example:

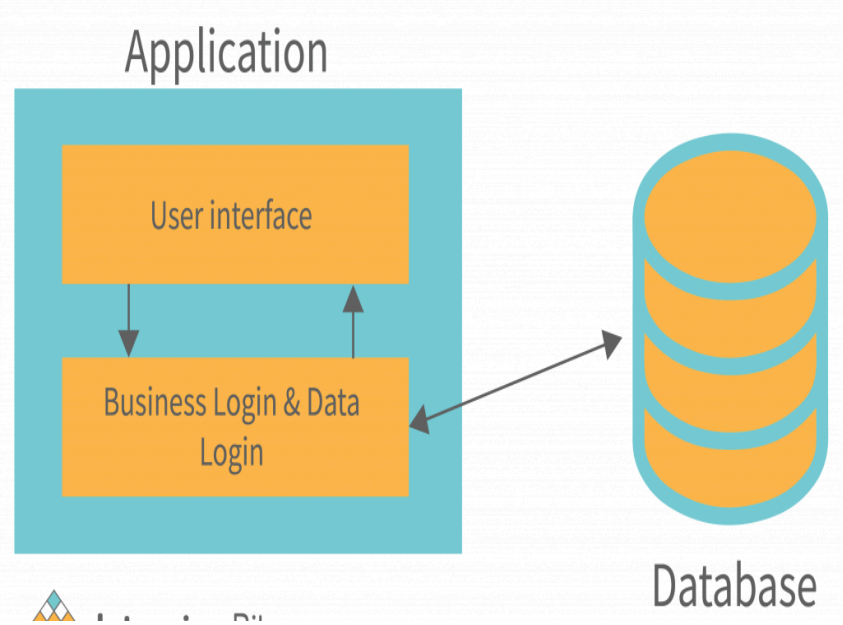
- A Java Swing application communicating with a MySQL database.

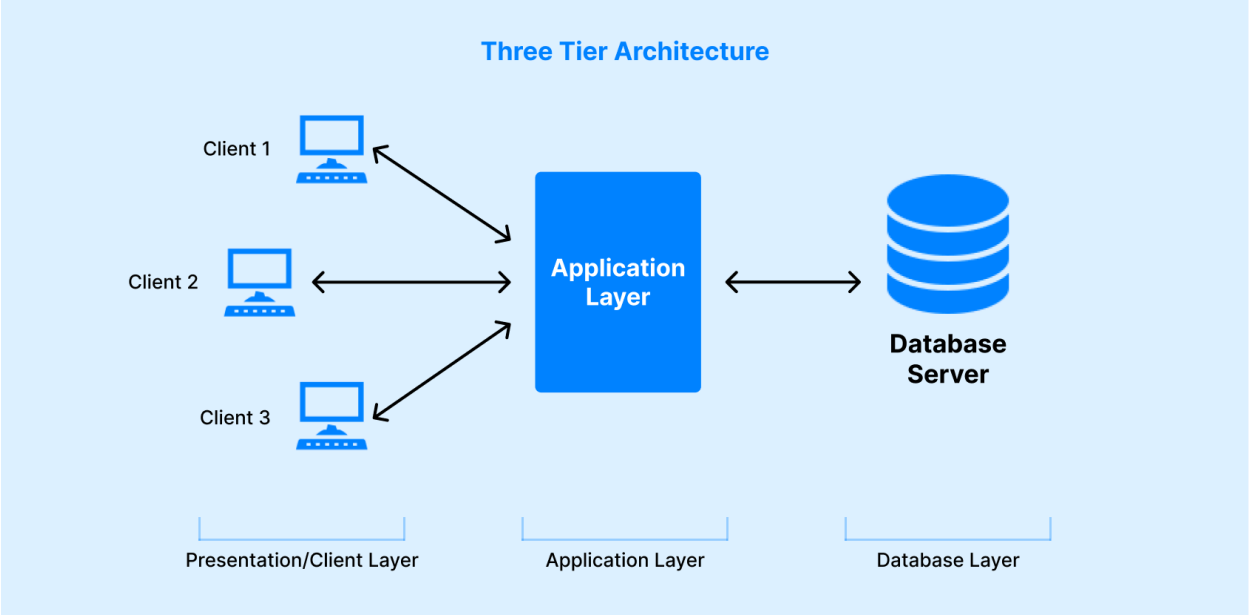
Advantages:

- Improved performance by splitting responsibilities.
- Easier to maintain than single-tier.

Disadvantages:

- Scalability is limited as the server becomes a bottleneck.
- Changes to business logic require redeployment of server code.





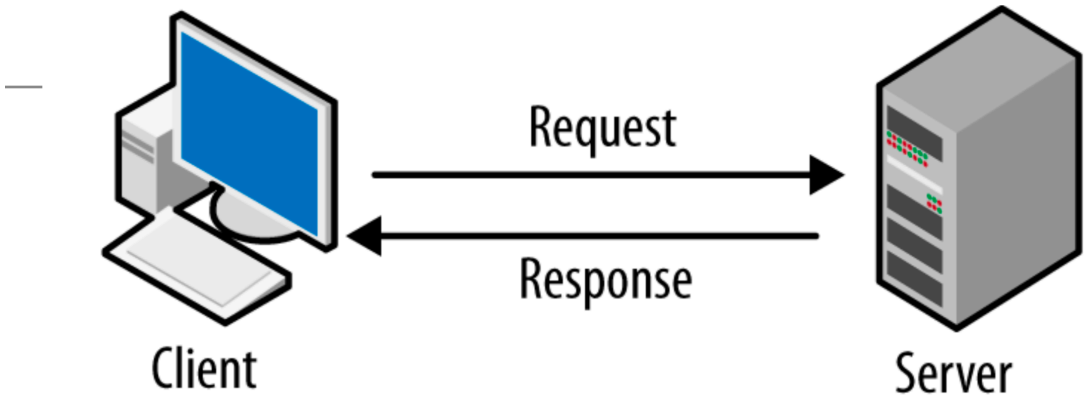
2. Communication Between Client and Server

2.1 How Communication Happens

- **Request-Response Model**: Clients send HTTP/HTTPS requests to servers, which respond with data or services.
- **Protocols**:
  - **HTTP/HTTPS**: Used for web pages and RESTful APIs.
  - **WebSockets**: For real-time communication.
  - **RPC (Remote Procedure Calls)**: For direct method invocation.

Example of HTTP Request-Response:

1. **Client**: Sends an HTTP GET request to `https://example.com/api/data`.
2. **Server**: Processes the request and responds with JSON data.



3.Monolithic vs. Microservices Architecture

3.1 Monolithic Architecture

In **monolithic architecture**, the entire application is built as a single unit. All functionalities, such as authentication, business logic, and database operations, are bundled together.

Advantages:

- Simple to develop and deploy.
- Easy to debug as the application is in one codebase.

Disadvantages:

- Difficult to scale specific parts of the application.
- Changes in one module may affect the entire system.
- Slower development for large teams due to dependencies.

Diagram:

```
Single Application -> Database
```

3.2 Microservices Architecture

In **microservices architecture**, the application is divided into smaller, loosely coupled services. Each service handles a specific functionality and communicates via APIs.

Advantages:

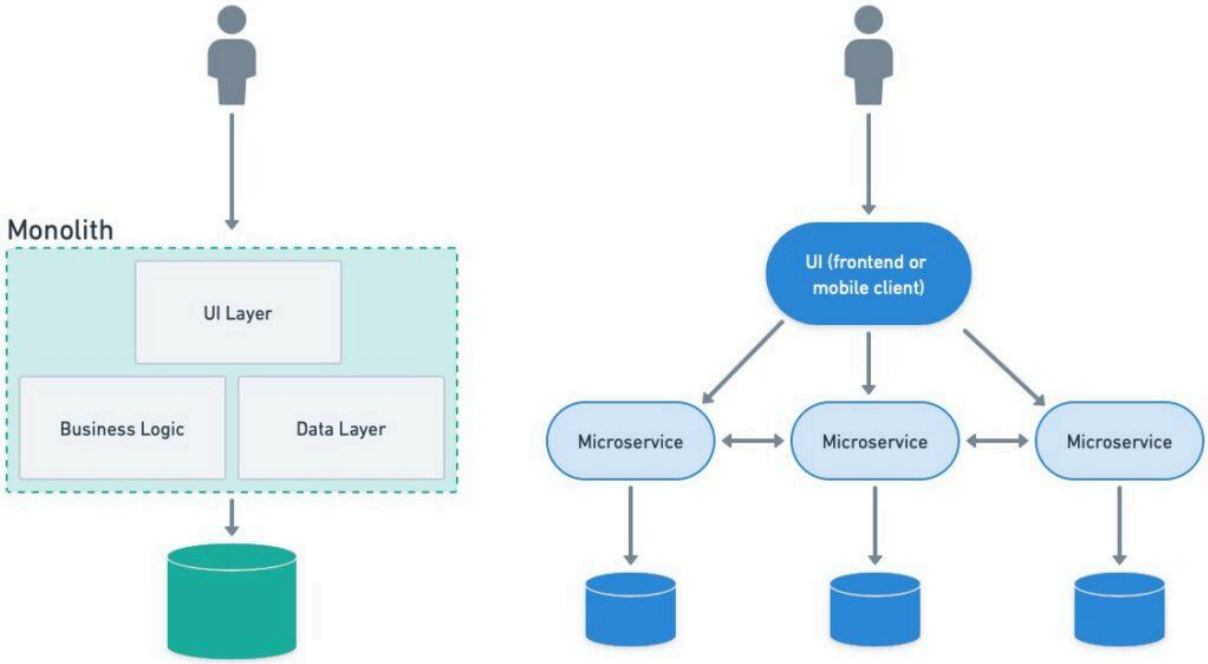
- Scalability: Services can scale independently.
- Flexibility: Easier to adopt new technologies for specific services.
- Faster development with small, autonomous teams.

Disadvantages:

- More complex to deploy and manage.
- Communication between services can become a bottleneck.

Diagram:

```
Service A <--> Database A
Service B <--> Database B
Service C <--> Database C
```



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4. What is a REST API?

4.1 Introduction to REST

REST (Representational State Transfer) is an architectural style for designing networked applications. It relies on stateless, client-server communication over HTTP.

Key Concepts:

1. **Resources**: Identified by URLs (e.g., `/users`, `/products`).
2. **HTTP Methods**:
    - **GET**: Retrieve data.
    - **POST**: Create new data.
    - **PUT**: Update existing data.
    - **DELETE**: Remove data.
3. **Statelessness**: Each request contains all the information needed to process it.
4. **Representation**: Data is usually represented in JSON or XML format.

Example REST API Endpoint:

- **Endpoint**: `https://example.com/api/users`

- **GET /api/users**: Fetch all users.
- **POST /api/users**: Create a new user.

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5. Introduction to Java Containers & Servlets

5.1 Java Containers

A **Java container** provides the runtime environment for Java web applications. It manages the lifecycle of servlets, handles requests, and ensures compliance with the Java Servlet Specification.

Examples:

- **Apache Tomcat**: A lightweight servlet container.
- **Jetty**: Used for embedded applications.
- **GlassFish**: A full-fledged Java EE application server.

5.2 Servlets

A **servlet** is a Java class used to handle HTTP requests and generate dynamic web content.

Example:

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.IOException;

public class HelloWorldServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,
HttpServletResponse response) throws IOException {
        response.setContentType("text/html");
        response.getWriter().println("<h1>Hello, World!</h1>");
    }
}
```

Lifecycle of a Servlet:

1. **Initialization**: `init()` method is called.
2. **Request Handling**: `service()` or `doGet()/doPost()` is invoked.
3. **Destruction**: `destroy()` method is called.

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6. Introduction to Software Design Patterns

Design patterns are reusable solutions to common software design problems. They are categorized into three main types:

6.1 Creational Patterns

Focus on object creation mechanisms.

Examples:

- **Singleton**: Ensures only one instance of a class.
- **Factory**: Creates objects without specifying the exact class.

6.2 Structural Patterns

Focus on organizing classes and objects to form larger structures.

Examples:

- **Adapter**: Converts the interface of a class into another.
- **Decorator**: Adds responsibilities to an object dynamically.

6.3 Behavioral Patterns

Focus on communication between objects.

Examples:

- **Observer**: Allows objects to subscribe to updates.
- **Strategy**: Encapsulates interchangeable behaviors.

Practical Example: Singleton

```
public class SingletonExample {
    private static SingletonExample instance;

    private SingletonExample() {}

    public static SingletonExample getInstance() {
        if (instance == null) {
            instance = new SingletonExample();
        }
        return instance;
    }
}

public class Main {
    public static void main(String[] args) {
        SingletonExample instance = SingletonExample.getInstance();
        System.out.println("Singleton instance: " + instance);
    }
}
```